

Similarly, the second inequality uses the boundedness of entries of Q -matrix; and $\dots$ represents the first two terms in a fold. Thus by Grönwall's inequality, it follows

$$\begin{aligned} \int_I |\hat{u}^{n,x}(s, \cdot) - \hat{u}^x(s, \cdot)| dx \leq & C \left(\int_I \int_s^T \left| q_{e,e'} \left(\hat{\phi}^x(t, e), \hat{Z}_{t,K}^x, \hat{Z}_{t,I}^x \right) - q_{e,e'} \left(\hat{\phi}^{n,x}(t, e), \hat{Z}_{t,K}^{n,x}, \hat{Z}_{t,I}^{n,x} \right) \right|^2 dt dx \right. \\ & \left. + \int_I \int_s^T \left| -f^x \left(t, e, \hat{Z}_{t,K}^{n,x}, \hat{Z}_{t,I}^{n,x}, \hat{\phi}^{n,x}(t, e) \right) + f^x \left(t, e, \hat{Z}_{t,K}^x, \hat{Z}_{t,I}^x, \hat{\phi}^x(t, e) \right) \right|^2 dt dx \right) \end{aligned} \quad (14)$$

Again the continuity of $(\hat{Z}_K, \hat{Z}_I)$ induces the convergence of the two q functions. This completes the proof.

Lemma 1.1: $\mathcal{Z}$ is closed in $C([0, T]; L^2(I; \mathbb{R} \otimes \mathbb{R}))$ equipped with norm $\|\cdot\|_{\mathcal{Z}}$.

Proof: First note $\mathcal{Z}$ is closed if and only if any convergent sequence $(f_n)_{n=1}^\infty$ in $\mathcal{Z}$ converges to some $f \in \mathcal{Z}$. For the sake of contradiction, suppose there exists $(f_n)_{n=1}^\infty \in \mathcal{Z}$ converges to $f \notin \mathcal{Z}$. Then there exists $t_0 \in [0, T]$ such that there exists a measurable set $\tilde{I} \subset I$ with $m(\tilde{I}) > 0$, that $f(t_0)(x) > \bar{A} + \epsilon_0$ or $f(t_0)(x) < -\bar{A} - \epsilon_0$ for some $\epsilon_0 > 0$, $x \in \tilde{I}$. Here m is a measure equipped on I . Then $\forall n \in \mathbb{N}$,

$$\int_I |f_n(t_0)(x) - f(t_0)(x)| dx \geq \int_{\tilde{I}} |f_n(t_0)(x) - f(t_0)(x)| dx > \epsilon_0 \cdot m(\tilde{I})$$

There is contradiction and thus completes the proof. ■